Project Title:
Analysis of Factors Influencing Individual Medical Insurance Charges

Stakeholder:
A health insurance company seeking to understand the key drivers of individual medical insurance costs.

Objective:
To identify which demographic and health-related factors (age, BMI, smoking status, number of children, and region) are most strongly associated with higher insurance charges.

Key Variables:
- Age
- Sex
- BMI
- Smoking status
- Number of children
- Region
- Insurance charges

Business Decision Supported:
This analysis will help the company:
- Adjust risk-based pricing models
- Identify high-cost customer segments
- Design preventive health incentives (e.g., smoking cessation programs)
- Improve premium structure fairness

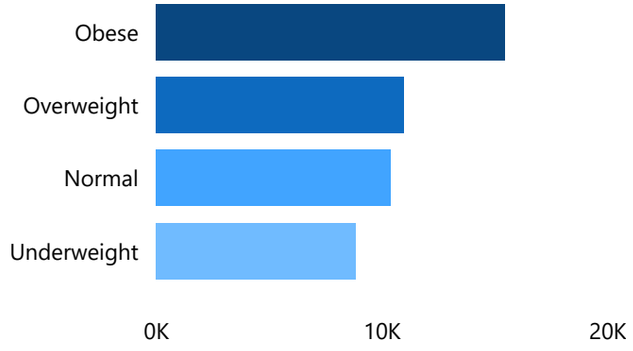
Medical Insurance Pricing Report

Gender
- [ ] female
- [ ] male

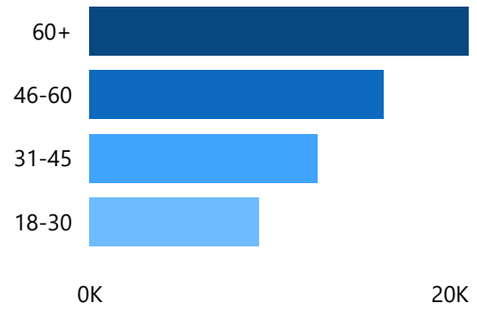
1338
Total Customers

13.27K
Average Charge

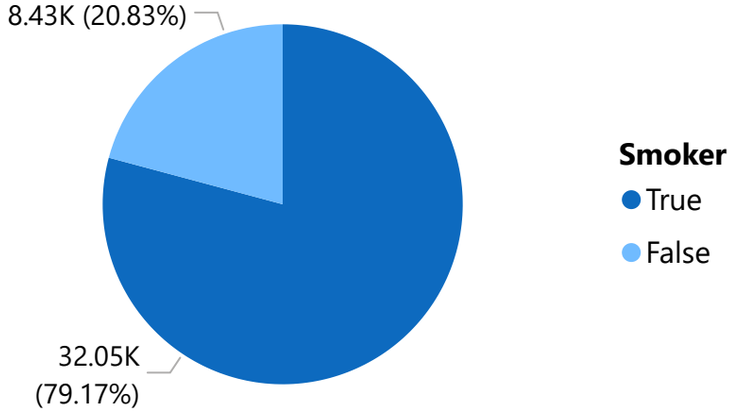
Charges by Body Mass Index



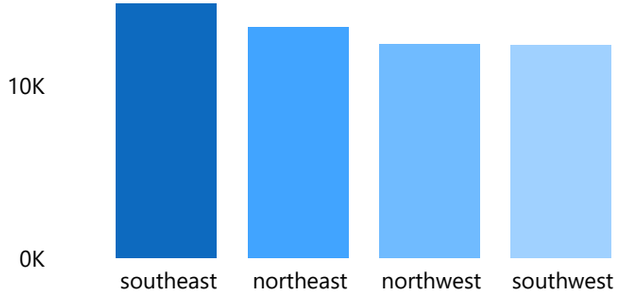
Charges by Age_group



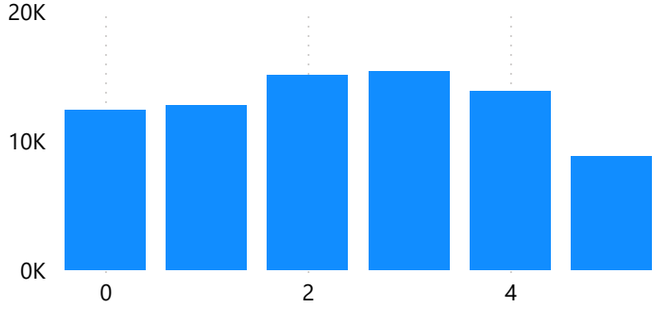
Charges by Smoking Status



Charges by Region



Charges by Number of Children



smoker	Sum of charges	Average of charges	Total Customers
False	8,974,061.48	8,434.27	1064
True	8,781,763.51	32,050.23	274
Total	**17,755,824.98**	**13,270.42**	**1338**

bmi_category	Sum of charges	Average of charges	Total Customers
Normal	2,304,248.94	10,379.50	222
Obese	11,129,796.29	15,479.55	719
Overweight	4,144,735.74	10,993.99	377
Underweight	177,044.01	8,852.20	20
Total	**17,755,824.98**	**13,270.42**	**1338**

age_group	Sum of charges	Average of charges	Total Customers
18-30	4,172,513.11	9,397.55	444
31-45	4,983,097.51	12,647.46	394
46-60	6,683,466.49	16,340.99	409
60+	1,916,747.87	21,063.16	91
Total	**17,755,824.98**	**13,270.42**	**1338**